

MMET 383 Lab Guide



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MMET 383 Lab

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Place: THOM 115C

Lab assignments needs to be submitted as a ".zip" file to Canvas;

If you finish your lab assignment during the lab, you may leave early;

A personal laptop is encouraged;

Attendance is required;

Office hour: 3-5pm Monday



You can get this slide
and the video tutorial
from this QR code:





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CS.Fall.2026

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New2026-383-Labs1-9.pdf

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Dr. Sheng-Jen (“Tony”) Hsieh

Lab #1 MMET383

Visual Basic Programming: Create a User Interface

Objective: Practice Visual Basic programming to develop a user interface using Tool Box.

Task #0: Listen to and take notes during the lab demonstration.

Task #1: Develop a user-interface using the Tool Box.

This interface will include **Radio Buttons, Image Boxes, and Text Boxes.**

Following are the requirements for the interface:

1. Three Radio buttons, Image boxes, and Text boxes are required within a form.
2. When one of the Radio buttons is pressed, its corresponding image and text box will be shown.
3. When Exit is pressed, the program ends.

Figure 1 shows the layout for the user interface.

Lab1-Task1



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<Hint>

Conversion: string <-> numeric.

e.g.

```
a = CInt(Textbox1.text)
```

```
Textbox2.text = CStr(b)
```

1 form, 5 textbox, 5 labels, 4 buttons;

Declare variables: **Increment_Result, Decrement_Result, Inc_Dec_Value, Count**

Initialize

-> Display the "Initial Value" as Decrement Result and Increment Result;

Increment

-> Run calculation:

Decrement_Result = Decrement_Result - Inc_Dec_Value;

Increment_Result = Increment_Result + Inc_Dec_Value;

Count calculation:

Count = Count + 1

Swap

-> Swap the current Increment Result and Decrement Result;

How to swap A and B:

Step 1: C = A

Step 2: A = B

Step 3: B = C

Initial: A = "coke", B = "tea", C = ""

Final: A = "tea", B = "coke", C = "coke"

A = "coke", B = "tea", C = "coke"

A = "tea", B = "tea", C = "coke"

A = "tea", B = "coke", C = "coke"

Clear

-> Clear all textbox entries and reset all variables;

Lab1-Task2



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<Hint>

- Textbox entry(Growth Rate (%)): 5
Rate -> 0.05
- Format the investment(2 decimal places):
e.g.
`Textbox1.text = Format(a, "$0.00")`

1 form, 5 buttons, 7 textbox and labels;

Declare variables:

Investor_Name	-> Name
Investment_Goal	-> FV
Years	-> Term
Growth_Rate	-> Rate
Initial_Investment	-> PV
Number_Of_Investment	-> Count
Total_Initial_Investment	-> Total_PV

Hide "Portfolio Summary":

Textbox and Label: Portfolio Summary, Investor Name,
Number of Investments, Total Initial Investment

Calculate

-> Get the variables

Calculation:

Initial Investment
Total Investment
Number of Investment

$PV = FV / (1 + Rate)^{Term}$
 $Total_PV = Total_PV + PV$
 $Count = Count + 1$

Clear Input

-> Clear the textbox entries of the current investment(goal, years, rate)
e.g. `Textbox.text = ""`

Club Summary

-> Display calculated total investment and number of investment in the
summary area
e.g. `Textbox.Visible = True`
`Label.Visible = True`

New Investor

-> Reset all textboxes and variables
e.g. `Textbox.text = ""`
Hide summary area
`Textbox.Visible = False`
`Label.Visible = False`

Exit

-> Exit the program using command "End"